

Math 504: Advanced Linear Algebra

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November 22, 2021 (unfinished!)

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Math 504 Lecture 22

1. Singular value decomposition

This will be just a brief introduction to singular value decomposition. For much more, see [Trefethen & Bau].

1.1. The $\text{Ker}(A^*A)$ lemma

Proposition 1.1.1 ($\text{Ker}(A^*A)$ lemma). Let $A \in \mathbb{C}^{m \times n}$ be any $m \times n$ -matrix with complex entries (not necessarily square).

Then:

(a) We have $\text{Ker } A = \text{Ker}(A^*A)$.

(b) We have $\text{rank } A = \text{rank}(A^*A)$.

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Proof. **(a)** Let $x \in \mathbb{C}^n$. Then, we have the following chain of equivalences:

$$\begin{aligned}
 (x \in \text{Ker } A) &\iff (Ax = 0) \\
 &\iff (\|Ax\| = 0) \quad (\text{since a vector is 0 if and only if its length is 0}) \\
 &\iff \left(\sqrt{(Ax)^* Ax} = 0 \right) \quad \left(\text{since } \|Ax\| = \sqrt{\langle Ax, Ax \rangle} = \sqrt{(Ax)^* Ax} \right) \\
 &\iff ((Ax)^* Ax = 0) \\
 &\iff (x^* A^* Ax = 0) \\
 &\iff (A^* Ax = 0) \\
 &\iff (x \in \text{Ker } (A^* A)).
 \end{aligned}$$

So we have shown that $(x \in \text{Ker } (A^* A)) \implies (x \in \text{Ker } A)$. But the converse implication is even easier:

$$(x \in \text{Ker } A) \iff (Ax = 0) \implies (A^* Ax = 0) \iff (x \in \text{Ker } (A^* A)).$$

So we get the equivalence

$$(x \in \text{Ker } A) \iff (x \in \text{Ker } (A^* A)).$$

Since this holds for all vectors $x \in \mathbb{C}^n$, we thus conclude that $\text{Ker } A = \text{Ker } (A^* A)$. This proves part **(a)**.

(b) The rank-nullity theorem yields

$$\begin{aligned}
 \text{rank } A &= n - \dim(\text{Ker } A) \quad \text{and} \\
 \text{rank } (A^* A) &= n - \dim(\text{Ker } (A^* A)).
 \end{aligned}$$

The right hand sides of these two equalities are equal (since part **(a)** yields $\text{Ker } A = \text{Ker } (A^* A)$). Thus, the left hand sides are also equal. In other words, $\text{rank } A = \text{rank } (A^* A)$. This proves part **(b)**. $\square$

Note that the above proposition really requires a matrix with complex entries. It cannot be generalized to matrices over an arbitrary field.

1.2. The singular value decomposition

Definition 1.2.1. Let A and B be two $m \times n$ -matrices with complex entries.

We say that A and B are **unitarily equivalent** if there exist unitary matrices $U \in \text{U}_m(\mathbb{C})$ and $V \in \text{U}_n(\mathbb{C})$ such that $A = UB V^*$.

Note the difference between “unitarily equivalent” and “unitarily similar”: The latter requires $A = UBU^*$, whereas the former only requires $A = UB V^*$.

Unitary equivalence is an equivalence relation (exercise!). A natural question is therefore: What is the “simplest” matrix in the equivalence class of a matrix?

Definition 1.2.2. A rectangular matrix $A \in \mathbb{F}^{n \times m}$ (for a field $\mathbb{F}$) is said to be **pseudodiagonal** if it satisfies

$$A_{i,j} = 0 \quad \text{whenever } i \neq j.$$

This is just the straightforward generalization of diagonal matrices to non-square matrices. In particular, a square matrix is pseudodiagonal if and only if it is diagonal. A pseudodiagonal 2×3 -matrix looks like this: $\begin{pmatrix} * & 0 & 0 \\ 0 & * & 0 \end{pmatrix}$. A pseudodiagonal 3×2 -matrix looks like this: $\begin{pmatrix} * & 0 \\ 0 & * \\ 0 & 0 \end{pmatrix}$. (Of course, any of the $*$ s can be a 0 too.)

Theorem 1.2.3 (SVD). Let $A \in \mathbb{C}^{m \times n}$. Then:

(a) There exist unitary matrices $U \in U_m(\mathbb{C})$ and $V \in U_n(\mathbb{C})$ and a pseudodiagonal matrix Σ such that

$$A = U\Sigma V^*.$$

In other words, A is unitarily equivalent to a pseudodiagonal matrix.

(b) The matrix Σ is unique up to permutation of its diagonal entries. (The matrices U and V are usually not unique.)

(c) Let $k = \text{rank } A$. Then, the matrix Σ has exactly k nonzero diagonal entries.

(d) Let $\sigma_1, \sigma_2, \dots, \sigma_n$ be the square roots of the eigenvalues of the Hermitian matrix A^*A , listed in decreasing order (so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_m$). Then, we have $\sigma_{k+1} = \sigma_{k+2} = \dots = \sigma_m = 0$, and we can take

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_k & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix} \in \mathbb{C}^{m \times n}$$

in part (a).

Definition 1.2.4. The triple (U, V, Σ) in part (a) of the theorem is called a **singular value decomposition** (short: **SVD**) of A . The numbers $\sigma_1, \sigma_2, \dots, \sigma_n$ are called the **singular values** of A .

Before we prove the theorem, a few words about the use of an SVD. Often in practice, you have a matrix A and you want to find a low-rank “approximation”

for A : that is, a matrix B that is “sufficiently close” to A and yet has low rank. Indeed, the idea for doing this is to compute an SVD of A – that is, $A = U\Sigma V^*$ with U, Σ, V as above – and then set all but the first few σ_i ’s to 0 in Σ .

Let us now prove the theorem.

Proof of the SVD theorem. Let $k = \text{rank } A$. Then, $k = \text{rank } (A^*A)$ (as we know from part **(b)** of the above proposition). Also, k is the rank of an $m \times n$ -matrix (namely, A), and thus satisfies $k \leq m$ and $k \leq n$.

The matrix A^*A is Hermitian, since $(A^*A)^* = A^* \underbrace{(A^*)^*}_{=A} = A^*A$. Moreover, this matrix A^*A is positive semidefinite, since each vector $x \in \mathbb{C}^n$ satisfies

$$\langle A^*Ax, x \rangle = \underbrace{x^*A^*}_{=(Ax)^*} Ax = (Ax)^* Ax = \|Ax\|^2 \geq 0.$$

The spectral theorem thus yields that A^*A can be written in the form

$$A^*A = V \cdot \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \cdot V^*,$$

where $V \in U_n(\mathbb{C})$ is a unitary matrix and where the $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A^*A listed in decreasing order. Consider this V and these $\lambda_1, \lambda_2, \dots, \lambda_n$. The eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ are nonnegative reals, since A^*A is positive semidefinite. Thus, we can set

$$\sigma_i := \sqrt{\lambda_i} \quad \text{for each } i \in [n].$$

Thus, $\sigma_1, \sigma_2, \dots, \sigma_n$ are nonnegative reals.

We have

$$\begin{aligned} k = \text{rank}(A^*A) &= \text{rank}(\text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)) && \left(\text{since } A^*A \stackrel{\text{us}}{\sim} \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \right) \\ &= (\text{the number of } i \in [n] \text{ such that } \lambda_i \neq 0) \\ &= (\text{the number of } i \in [n] \text{ such that } \sigma_i \neq 0) && \left(\text{since } \sigma_i = \sqrt{\lambda_i} \right). \end{aligned}$$

Hence, exactly k of the numbers $\sigma_1, \sigma_2, \dots, \sigma_n$ are nonzero. Since $\sigma_1, \sigma_2, \dots, \sigma_n$ are nonnegative reals and sorted in weakly decreasing order, this entails that

$$\begin{aligned} \sigma_1 &\geq \sigma_2 \geq \dots \geq \sigma_k > 0 && \text{and} \\ \sigma_{k+1} &= \sigma_{k+2} = \dots = \sigma_n = 0. \end{aligned}$$

Let $v_1, v_2, \dots, v_n$ be the columns of the unitary matrix V . Thus, $(v_1, v_2, \dots, v_n)$ is an orthonormal basis of $\mathbb{C}^n$ (since V is unitary). Since V is the unitary matrix in a spectral decomposition of A^*A , the columns of V are eigenvectors of A^*A , corresponding to the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. In other words,

$$A^*Av_i = \lambda_i v_i \quad \text{for each } i \in [n].$$

For each $j \in [k]$, set

$$u_j := \frac{1}{\sigma_j} Av_j.$$

(The division by σ_j is well-defined, since $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$ imply $\sigma_j \neq 0$.) We claim that the tuple $(u_1, u_2, \dots, u_k)$ is orthonormal. Indeed:

- For any $i \neq j$, we have

$$\begin{aligned} \langle u_i, u_j \rangle &= \left\langle \frac{1}{\sigma_i} Av_i, \frac{1}{\sigma_j} Av_j \right\rangle = \frac{1}{\sigma_i \sigma_j} \langle Av_i, Av_j \rangle = \frac{1}{\sigma_i \sigma_j} (Av_j)^* Av_i \\ &= \frac{1}{\sigma_i \sigma_j} v_j^* \underbrace{A^* Av_i}_{=\lambda_i v_i} = \frac{1}{\sigma_i \sigma_j} v_j^* \lambda_i v_i = \frac{\lambda_i}{\sigma_i \sigma_j} \underbrace{v_j^* v_i}_{=\langle v_i, v_j \rangle} = 0, \end{aligned}$$

(since $(v_1, v_2, \dots, v_n)$ is orthonormal)

and thus $u_i \perp u_j$.

- For any i , we have

$$\begin{aligned} \langle u_i, u_i \rangle &= \left\langle \frac{1}{\sigma_i} Av_i, \frac{1}{\sigma_i} Av_i \right\rangle = \frac{1}{\sigma_i \sigma_i} \langle Av_i, Av_i \rangle = \frac{1}{\sigma_i \sigma_i} (Av_i)^* Av_i \\ &= \frac{1}{\sigma_i \sigma_i} v_i^* \underbrace{A^* Av_i}_{=\lambda_i v_i} = \frac{\lambda_i}{\underbrace{\sigma_i \sigma_i}_{=1}} v_i^* v_i = v_i^* v_i = \langle v_i, v_i \rangle = 1 \end{aligned}$$

(since $\sigma_i = \sqrt{\lambda_i}$)

(since $(v_1, v_2, \dots, v_n)$ is orthonormal), and thus $\|u_i\| = 1$.

So $(u_1, u_2, \dots, u_k)$ is an orthonormal tuple of vectors in $\mathbb{C}^m$. As we know, we can extend it to an orthonormal basis $(u_1, u_2, \dots, u_m)$ of $\mathbb{C}^m$. So let us do this. Let U be the $m \times m$ -matrix with columns $u_1, u_2, \dots, u_m$. Thus, U is a unitary matrix, since its columns form an orthonormal basis of $\mathbb{C}^m$.

Set $\sigma_i := 0$ for every $i > n$. Note that $\sigma_i = 0$ for every $i > k$, since we already have $\sigma_{k+1} = \sigma_{k+2} = \dots = \sigma_n = 0$.

Let

$$\Sigma := \begin{pmatrix} \sigma_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_k & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix} \in \mathbb{C}^{m \times n}.$$

Now, we claim that $A = U\Sigma V^*$. To prove this, we shall show the equivalent claim $AV = U\Sigma$.

It is sufficient to prove that

$$(AV)_{\bullet,j} = (U\Sigma)_{\bullet,j} \quad \text{for each } j \in [n]$$

(recall that the notation $M_{\bullet,j}$ means the j -th column of a matrix M). So let us fix $j \in [n]$ and try to prove that $(AV)_{\bullet,j} = (U\Sigma)_{\bullet,j}$.

- If $j \leq k$, then we have

$$\begin{aligned} (AV)_{\bullet,j} &= A \underbrace{V_{\bullet,j}}_{=v_j} = Av_j = \sigma_j \underbrace{u_j}_{=U_{\bullet,j}} \quad \left(\text{since } u_j = \frac{1}{\sigma_j} Av_j \right) \\ &= \sigma_j U_{\bullet,j} = (U\Sigma)_{\bullet,j}. \end{aligned}$$

- If $j > k$, then we can apply the $A^*Av_i = \lambda_i v_i$ equality to $i = j$ and obtain

$$A^*Av_j = \underbrace{\lambda_j}_{=0 \text{ (since } j > k)} v_j = 0,$$

so that $v_j \in \text{Ker}(A^*A) = \text{Ker } A$ (by the proposition above); now,

$$(AV)_{\bullet,j} = A \underbrace{V_{\bullet,j}}_{=v_j} = Av_j = 0 \quad (\text{since } v_j \in \text{Ker } A),$$

but also

$$(U\Sigma)_{\bullet,j} = \underbrace{\sigma_j}_{=\sqrt{\lambda_j}=\sqrt{0}=0} U_{\bullet,j} = 0,$$

and therefore $(AV)_{\bullet,j} = 0 = (U\Sigma)_{\bullet,j}$.

Thus, we have proved $(AV)_{\bullet,j} = (U\Sigma)_{\bullet,j}$ in both cases, and this completes the proof of $AV = U\Sigma$. Therefore,

$$\underbrace{U\Sigma}_{=AV} V^* = A \underbrace{VV^*}_{=I_n \text{ (since } V \text{ is unitary)}} = A,$$

so that $A = U\Sigma V^*$, as desired.

This proves parts **(a)**, **(c)** and **(d)** of the theorem.

(b) We must show that if P is a pseudodiagonal matrix such that A is unitarily equivalent to P , then P and Σ have the same diagonal entries up to order.

Rough outline: The singular values of a matrix are invariant under unitary equivalence. In other words, if $A = UBV^*$ for some unitary U and V , then A and B have the same singular values. (In fact, $A = UBV^*$ entails

$$A^*A = (UBV^*)^*UBV^* = VB^*\underbrace{U^*U}_{=I_m}BV^* = VB^*BV^* \stackrel{\text{us}}{\sim} B^*B;$$

this shows that A^*A and B^*B have the same eigenvalues; therefore A and B have the same singular values.) Thus, P and Σ must have the same singular values (since P and Σ are both unitarily equivalent to A). However, for a pseudodiagonal matrix, the singular values are simply the diagonal entries. (Fine print: There might be 0s not accounted for.) $\square$